

In Yellowstone, Migratory Bison Reawaken a Landscape [C1]

Secondo un recente studio condotto a Yellowstone, il reinserimento dei bisonti apporterebbe numerosi benefici all'intero di un ecosistema. Oggi, il parco ospita l'ultima mandria migratoria rimasta.

Tens of millions of bison once [roamed](#) North America, [grazing](#) on grasslands, in forests and on plains, from the Great [Basin](#) of Nevada to the Atlantic coast. It is difficult today to imagine the size of [herds](#) that are no longer seen; Lakota oral histories [gauged](#) herd size by the number of days it took them to pass. Likewise, studying the role that migratory bison played in those ecosystems has become nearly impossible. After being driven nearly to extinction in the 1800s, the animals now exist only in small [herds](#). But a new study conducted in Yellowstone National Park, where the last migratory herd still roams, offers a [glimpse](#) into the crucial role that these animals once played in restoring their ecosystem and perhaps still could. "If we value a system, we need to allow them to operate as close to naturally as possible," said Bill Hamilton, an ecologist at Washington and Lee University and an author of the study. "And this was a great case in point of how that can work." Of the roughly 400,000 [extant](#) bison, more than five thousand live in Yellowstone's 3,500 square miles [906,495 hectares]; there are two [herds](#) in the park, one of which is migratory. (Most of the other bison outside the park are held in privately- owned [herds](#).) The migratory bison of Yellowstone travel more than one thousand miles in a year, [grazing](#) different habitats along a fifty-mile migratory route in the northern ecosystem. A 19th-century image shows a Native American on horseback hunting a bison, on which they historically rely for food, shelter, clothing, and spiritual practices. The study, published recently in the journal Science, examined how bison changed the [soil](#) and vegetation along their migratory route. [Outwardly](#) the effect can look like [overgrazing](#). But the researchers found that bison essentially allow plants to keep growing: by [grazing](#) and moving on, the animals increase the density of microbes and nitrogen, an essential chemical for plant growth, in the [soil](#), improving the nutrition for herbivores by up to 150 per cent in some areas. Troy Heinert, a member of the Rosebud Sioux

tribe and chief of the branch of bison management for the Bureau of Indian Affairs, said that such research validated what Indigenous peoples have known for generations. "Buffalo helped shape this continent," Heinert said. "And the more buffalo that are out there, the ecosystems are improved for all other animals as well." Current Affairs

Can the Dodo Come Back to Life? Secondo Colossal Biosciences, tra circa un decennio migliaia di dodo potrebbero tornare ad abitare Mauritius. Si tratterebbe di una grande impresa scientifica o di un azzardo morale? Daniel Francis Yellowstone, established as a national park in 1872, provides scientists with a rare opportunity to study how large herbivores have shaped [grazing](#) patterns across decades. Predators and herbivores, including wolves and bison, have been gradually [restored](#) to the landscape. "People look to Yellowstone and the research that comes out of Yellowstone to be some of the most comprehensive understanding that we have of a site," said Sally Koerner, a community ecologist at University of North Carolina at Greensboro, who was not involved in the research. From 2015 to 2022, Hamilton and his team took monthly samples of [soil](#) from sixteen sites, from [valley bottoms](#) to wet, high-elevation habitats. At each site, the researchers compared natural [grazing](#), where bison were the primary herbivores, with [grazing](#)-exclusion sites. They found that bison helped plants [grow steadily](#) by increasing microbial abundance and the rate of decomposition, resulting in a greater diversity of plants across the migratory landscape. When given more land to roam, bison create a mix of different habitats and improve the quality of the [soil](#). The power of large, migrating herbivores [lies](#) not only in their physical size but also in their numbers, density and freedom to migrate, the researchers noted. "This kind of serves as an example of how... if large numbers could get large areas of land, what it might look like for restoring bison elsewhere," Hamilton said. Bison graze among the hot springs and geysers of Yellowstone. Doug Frank, an ecosystem ecologist at Syracuse University who led many early studies of bison [grazing](#) in Yellowstone, described the new results as "very solid." The effort that went into it "was [stunning](#)," he added. "They're studying these [grazing](#) patterns across a vast ecosystem where the animals can migrate freely," he said. For many years, Native American tribes have sought to restore bison to their lands. Conservation biologists and tribal leaders have

discussed the possibility of removing park [fences](#) to allow migration beyond the official borders. But, Koerner wondered, how big does that migration need to be to benefit the ecosystem? And how big can it practicably be? “That’s going to take hundreds of studies to try and figure out and [tease this apart](#) at a smaller, [finer, grain scale](#),” Koerner said. “Because the whole world can’t function like Yellowstone.” Hamilton noted that the study confirmed Indigenous people’s traditional knowledge of the value of the bison. “They didn’t need science to demonstrate that,” he said. “But in modern times, other people do, and I find that as a great outcome.” The study’s [findings](#), he added, were “not just cool ecology.” Heinert [concurred](#). “The connection between Indigenous people and buffalo has never been broken,” he said. “But the restoration is extremely important, so that our younger generations know what having buffalo on the landscape means.” Published in The New York Times on 3 September 2025. Reprinted with permission.

Glossary

- **Basin** = bacino
- **glimpse** = sguardo
- **restored** = reintrodurre
- **fences** = recinzioni
- **tease this apart** = scomporre, analizzare nel dettaglio
- **finer, grain scale** = scala ad alta definizione
- **gauged** = calcolare
- **overgrazing** = sovrapascolo
- **grazing** = pascolare
- **herds** = mandrie
- **soil** = suolo, terreno
- **Outwardly** = dall'esterno
- **lies** = risiedere in
- **stunning** = impressionante
- **findings** = risultati
- **roamed** = girovagare
- **extant** = esistenti
- **valley bottoms** = fondivalle
- **grow steadily** = crescere in modo costante
- **concurred** = concordare, essere d'accordo